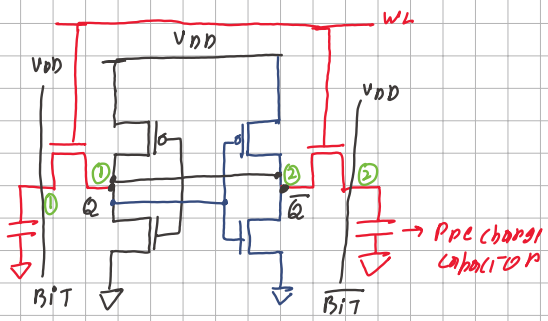

JNOTE.
JNOTE.
JNOTE.



Read

1. Let $Q=1, \bar{Q}=0$ $Q=0$ $\bar{Q}=1$

2. $WL=1$ (access transistor on)
   we can access data

3. Bit & $\overline{BIT}$ => o/p

4. initially both pre charge capacitors are charged => $C = V_{DD}$

5. AT node 1 $Q=1$ $V_1=1$ $Q=0$ $V_1=1$
   it will remain same Bit will ↓
   $BIT=1$

   AT node (2) $\bar{Q}=0$ $V_2=1$ $\bar{Q}=1$ n channel

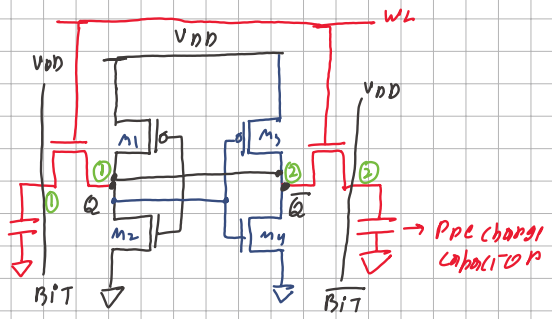
   => Pre charge capacitor will discharge

   => $\overline{BIT}$ will ↓ bit will ↓

6. Bit & $\overline{BIT}$ sent to sense amplifier

   => sense amplifier will say → bit ↓ & $\overline{BIT}$ ↑

   If Bit = const & $\overline{BIT}$ dec
   => o/p = 1 o/p = 0



write

1) $Q = 0$ $\bar{Q} = 1$ initially
2) $WL = 1$
3) BIT & $\overline{BIT} = 1/b$
   we make
4) $\overline{BIT} = GND$
5) at ② V_C dec, & block in V CMOS will get affected.
   at $\bar{Q} < V_{Tn}$ => M_2 off M_1 on

=> o/b $Q = 1$

Read

BIT $Q = 0$ & $\bar{Q} = 1$

V_1 should not exceed V_{Tn} of M_4